

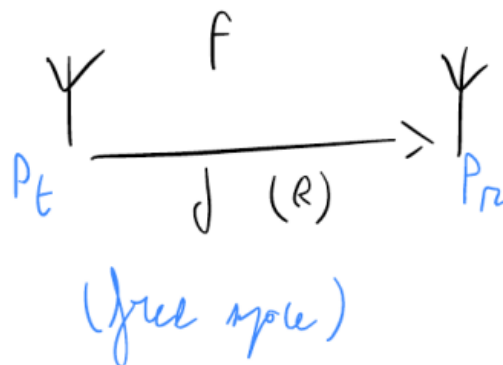
Capitolo 10

Lesson 5-12-2025

10.1 Channel modeling

10.1.1 Friis' equation and discussion on the choice of the transmission frequency

Let's say we have



Which parameters do we need? The gains of the transmit and receive antennas: G_T, G_R .

Friis' equation

What's the equation which binds the two?

$$P_r = P_t G_t G_r \frac{1}{d^2 f^2} \left(\frac{c}{4\pi} \right)^2 = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 ;$$

this is Friis' equation; what does this tell us?

We can't do anything on the distance! **We can do something on the frequency** though; if we double the frequency, the received power will be divided by 4. The lower frequencies will be the best ones. Of course we can't go baseband: the size of the

antenna is somehow proportional to λ ; if we have $f = 0$, we'll need to have $\lambda = \infty$ and thus we'd need an extremely large antenna.

In Europe the unlicensed band is around **868MHz**. We then have other ones at **2.4, 5GHz**: those are used by the wifi. If we have a router and home, let's imagine we have a switch between the two; normally it transmits at both. When we transmit, we can decide which to use; we might think we'll certainly choose 2.4GHz, since it will get 4 times the power the other gives! There are other considerations; speed is something we want: the bitrate does not only depend on the power, but also on the bandwidth. Here the bandwidth is larger. At 5GHz we have **higher bitrate but more attenuation**, while at 2.4GHz we don't have as much attenuation but we also have a lower bandwidth.

If we consider the channel itself, it would prefer lower frequencies. It's also true that **sometimes we want attenuation** (voice is attenuated: we can talk without disturbing the neighbour by lowering the voice).

We don't want the router of our neighbour to enter our house, as that would create interference. The fact that the waves are attenuating is good, if we look at that kind of principle.

Here the exponent is 2. We know that now wifi is converging towards mmwaves; these are frequencies for which the λ is comparable with **1mm**, meaning that the frequency is $f > 6\text{GHz}$. Why so? mmwaves have some other problems; if we are in the air (not in the free space anymore), we have oxygen, water, ...If we take a frequency which is resonating certain molecules, then we have peaks of absorption at certain frequencies. The frequencies at mmwaves experience the phenomenon of **blockage**. If the wave just meets a leaf, it's immediately blocked. Such high frequencies (like mmwaves) are blocked by many types of obstacles. Point is that higher frequencies have very large and non-crowded bandwidth: if we're in line of sight, we can upload gigabits per second!

Of course the more bandwidth we have, the larger the bitrate we can achieve (this follows easily from $R/B = C$: for a fixed capacity, increasing the bandwidth gives us a better bitrate). Let's suppose we can get the line of sight.

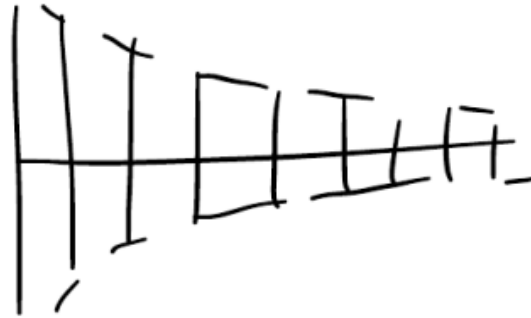
If we consider that each cellular phone has a bandwidth of 2MHz, this means that

$$\frac{60\text{GHz}}{2\text{MHz}} = 3 \cdot 10^4$$

(where 60GHz is the frequency of one of the more recent standards) this is a really big attenuation factors; how can we do this, then? 5G is using mmwaves; 6G is thinking of THz! We solve the problem of the high attenuation by working on the gain of antennas; amplifiers are not the way, as they introduce noise; the antennas are passive elements.

Since we have mmwaves, we can make an antenna which is just of the size of a millimeter; we can thus have multiple small antennas in a very small area. Going back again: we can set the gains really high. This is like directing the power specifically in a certain direction.

The gain comes from the fact that the signal is propagating specifically in one direction.



“ In-class stuff

THz is really horrible; generally telecommunications have always been like: “we make this, somebody will use it”; now telecommunications has a market pull: they say *we need something* and we’re the ones finding such things. The use case for THz waves is to replace the cables in data centers. This would allow for great simplifications in situations with many cables.

Shows a picture with attenuation depending on the transmission frequency. The spots of minima are candidates for 5G; wifi instead uses the spots where attenuation is higher; this is because wifi knows it does not need to go far in distance. The maximas are used depending on the application; if we just need low distance transmission, we can use the maxima as well.

Now we should have an idea of how the frequencies are selected. There is a lot of discussion about the human damages which according to certain people are being caused by electromagnetic waves.

Controversy of people not wanting 5G; we have a parameter called SAR (specific absorption rate): we can’t commercialize stuff which has wrong values of this. In principle, high frequencies on molecules should not be more dangerous than the lower ones; actually, the low frequencies are even worse.

10.2 Modeling approaches

10.2.1 Modeling of radio propagation (LOS, NLOS, effects on the signals)

Handwritten note: Two approaches → deterministic → ray tracing (difficult) (GPU). Stochastic → lights... used for underwater communications.

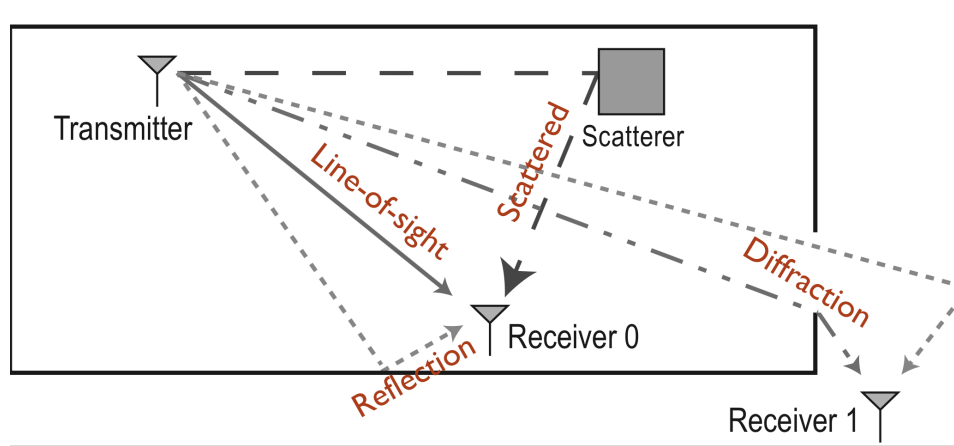
Highlights; we’ll skip them.

Propagation modeling; there are different effects that we should take into account. The aim is to develop a mathematical model of what happens physically!

The best that can happen is that our signal directly reaches the receiver, through a line of sight (LOS); generally, what happens is that the signal actually reaches the receiver

through some different type of path, aka a non-line of sight (NLOS). We have different possibilities, when it comes to NLOS:

1. A wave might bounce off a smooth object of size exceedingly large relative to the wavelength of the signal: we have **reflection**; each incoming wave is mapped onto a single reflected direction.
2. **Scattering** is said to have occurred when a wave travels through a medium that contains objects whose dimensions are smaller than or comparable to the wavelength, and where the number of such objects is large. Then, each incoming wave is mapped onto many scattered ones.
3. Radio waves also bend around sharp edges, a phenomenon that is referred to as **diffraction** and that plays a pivotal role in urban settings. Indeed, a major contributor to the propagation of radio signals in such settings is rooftop diffraction.



Ray tracing example: Reflection, Scattering, Diffraction

The Maxwell equations are not enough to describe the various effects we can have, such as diffraction, reflection, scattering. Only if we're lucky do we have a line of sight. In order to solve this we'd need to solve differential equations which are exceedingly more complicated: we could not get out of that, not even with super computers.

Clutter

In an urban scenario, we refer to the **propagation clutter** as the set of objects which cause the NLOS effects; we can have either an elevated transceiver (if it's elevated with respect to the clutter) or a *within clutter* transceiver.

We'll then derive an abstraction simple enough for us to be able to handle the model, but also complicated enough to be able to capture the major effects which are happening. Such a model is key of information engineering. We always need to be careful about the conditions which allowed us to form the model.

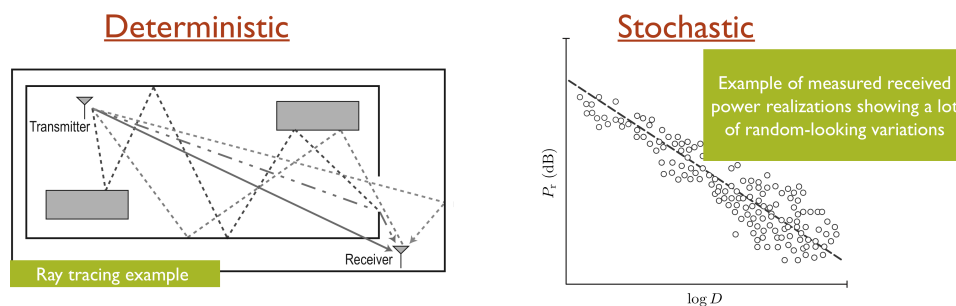
It's difficult (impossible!) that things go better than what is predicted by the Friis equations, which only work in line of sight. We'll then develop a mathematical description of multi-path propagation channels inspired by propagation channel measurements.

Sensitivity of the receiver

Sensitivity of the receiver: the minimum amount of power needed to correctly receive a signal with a certain probability. Sensitivity of -100dBm; if we receive any greater power, we'll receive the signal and interpret it correctly with a good probability. We have very expensive softwares able to map a whole area this way. It's also true that "high power signal" does not mean that we have a clean signal.

10.2.2 Modeling approaches (deterministic, stochastic); elements of the modelization

We know that in the end we won't generally develop a deterministic model, as that would be extremely complicated (involving ray tracing and such); we only do this sometimes, like with millimeter waves (since they generally require ray tracing of indoor sites). More often, we do stochastic models!



Deterministic vs Stochastic approaches

Channel modeling (pp. 131-208), page 4

In general, a stochastic model is embodied by the **conditional distribution of the channel output given its input**. When the channel is linear, as is the case in the wireless medium, this is tantamount (*equivalente*) to a stochastic impulse response.

A critical distinction which needs to be made when modeling is that between large-scale and small-scale phenomena.

Channel modeling (pp. 131-208), page 4

Specifically, what is **assumed** is that the **distribution of the channel response is locally stationary within certain neighborhoods around transmitter and receiver**.

We'll just assume wide-sense stationarity, but we might as well assume strict-sense stationarity. The neighborhoods in which the signal can be considered to be stationary is generally in the order of tens to hundreds of wavelengths.

☰ Channel modeling (pp. 131-208), page 4

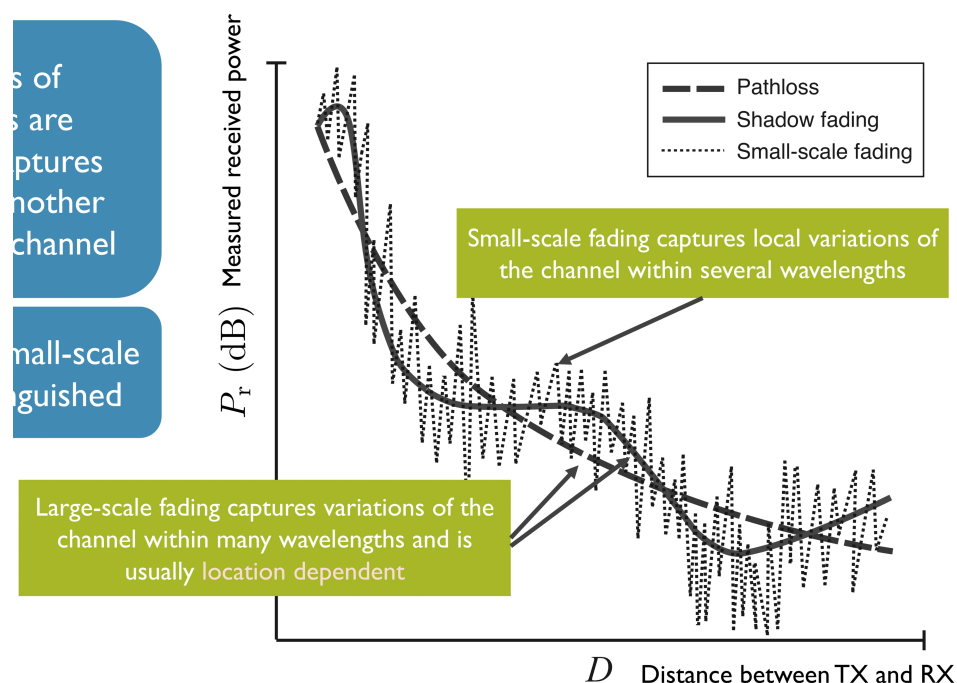
Large-scale phenomena impact the local distribution to which the small-scale phenomena conform, **generally the local-average power**.

☰ Channel modeling (pp. 131-208), page 5

Because of the importance of the decoupling of large- and small-scale phenomena, we use distinct variables to model them and carry those throughout the derivations in the entire book. While this takes a small toll in terms of the compactness of some expressions, it reinforces the separation between quantities that change at very different scales in time and frequency, and which have markedly different effects on the communication.

Just an insight on the approach we'll further discuss later: we generally draw a regression line trying to minimize the distance between the line and the various values. We make great campaigns of measurements, in order to obtain sufficient data to model this. We then decompose everything in two (+1) components:

1. **Large-scale fading:** if we go around the corner, we have another propagation, changing abruptly; this depends heavily on where we are! This change is not smooth.
2. **Small-scale fading:** depends on how the various waves come to the receiver and how they arrive at the receiver; if we move our phone up and down, we'll have variations in the signal.
3. **Line loss:** this changes depending on what we have in-between. There are parameters to tune if the loss is due to walls, ceilings, iron, ...; this needs to be managed carefully.



Decoupling of fading types over distance

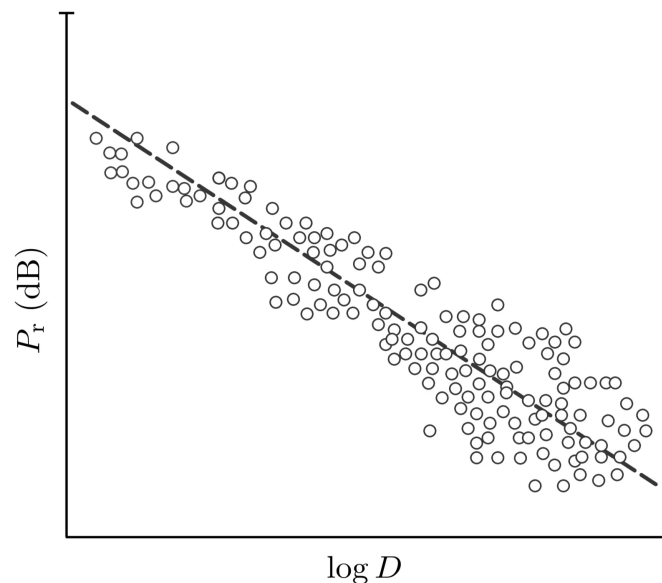
In MIMO, we want to have full-rank matrices; small scale fading can help to make the various matrices full rank!

10.3 Large scale phenomena

10.3.1 Pathloss and shadow fading (their separation, distribution of the shadow fading), pathloss; large-scale channel gain G and partial values P_t, P_r, G_t, G_r

 Channel modeling (pp. 131-208), page 5

Under the separation argued in the previous paragraph, large-scale phenomena are those that determine the local distribution of the stochastic channel response. The prime feature of that distribution is, of course, the power. If one plots, for a given transmit power, the local-average received power (in log-scale) obtained from many empirical measurements in otherwise identical conditions as a function of the distance (also in log-scale) between transmitter and receiver, the result is invariably a cloud of scattered points, such as the one in the figure below



The line we see in the figure above is a linear regression line which relates the local-average received power to the distance. Such a law represents what we call the **pathloss**, which we'll denote as L_p ; the scattered measurements around the law is what we call **shadow fading**, denoted by χ . The pathloss and the shadow fading put together give the large-scale description of virtually every existing channel model.

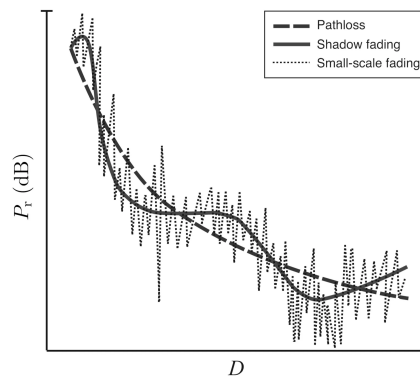
 Channel modeling (pp. 131-208), page 6

Although recent work indicates, on solid analytical footings based on random walks and diffusion theory, that nonlinear regressions may sometimes be superior [203, 204], the **combination of pathloss and shadow fading suffices for our purposes**.

 Channel modeling (pp. 131-208), page 6

The pathloss can be interpreted as the expected attenuation over an ensemble of realizations of a given propagation scenario, whereas the shadow fading individualizes, through randomization, the attenuation of each such potential realization.

Another common interpretation of the large-scale phenomena is that the **pathloss quantifies the local-average attenuation** tied exclusively to distance whereas the **shadow fading accounts for the presence of obstacles** that cause the local-average attenuation to deviate from the value predicted by the pathloss. This is illustrated in Fig. 3.4 (below here), which depicts how both pathloss and shadow fading affect the evolution of the received signal power as the distance between transmitter and receiver increases over a generic trajectory; the pathloss is monotonic, whereas the shadow fading can oscillate as obstacles come in and out of the way. It was suggested in [205] that **the shadow fading attenuation, χ , could follow a lognormal distribution** and empirical measurements have repeatedly confirmed the excellent fit to data offered by such distribution [206, 207]. The most widely accepted justification for this log-normal nature is that shadow fading represents the product of the losses introduced by the several obstacles traversed by the signal and, in log-scale, such a product becomes a sum of independent terms. Despite the fact that those terms can hardly be assumed to conform to the same distribution, and that the number of such terms may not be particularly large, the robustness of the central limit theorem (see Appendix C.1.10) renders the sum surprisingly well modeled by a Gaussian random variable.



Received signal power as a function of distance, illustrating the large-scale (pathloss and shadow fading) and the small-scale propagation phenomena.

Figura 10.1: Received signal power as a function of distance, illustrating the large-scale (pathloss and shadow fading) and the small-scale propagation phenomena. Fig. 3.4

We thus have that under the log-normal model, $\chi|_{\text{dB}}$ is Gaussian with PDF

$$f_{\chi|_{\text{dB}}}(\xi) = \frac{1}{\sqrt{2\pi}\sigma_{\text{dB}}} \exp\left(-\frac{\xi^2}{2\sigma_{\text{dB}}^2}\right), \quad (3.1)$$

where σ_{dB} is the standard deviation which is generally assumed to be independent of the distance for the sake of simplicity.

 *Channel modeling (pp. 131-208), page 7*

Outdoors, σ_{dB} tends to range between 8 and 12 dB. Indoors, it is usually smaller.

Log-normal random variable is like a Gaussian in the log domain! We use a Gaussian because it is given by the summing of various different random variables.

The signal was represented as zero-mean: we can easily include a mean in the log-scale Pathloss. We can also rewrite the distribution in linear scale, see [08.3_pp_131_208_Channel_modeling page 7](#).

Channel modeling (pp. 131-208), page 7

A relevant property confirmed through empirical observations is that, as intuition would have it, shadow fading **can be broken up into distinct pieces: one** that depends on the **location of the transmitter** and **one** that depends on the **location of the receiver** [207]. Two mutually distant receivers communicating with the same transmitter (or two mutually distant transmitters communicating with the same receiver) experience the same shadow fading for one piece but a completely different shadow fading for the other. These two pieces, whose values add up in log-scale, tend to have similar strengths. Furthermore, measurements reported in [208] point to an exponential model for the spatial autocorrelation of each piece, with a correlation distance that is on the order of tens of meters outdoors, depending on the environment, and substantially shorter indoors [209]. For more refinements on the modeling of shadow fading, the reader is referred to [210, 211] and references therein.

We can express the local-average received power P_r as a balance of gains and losses according to the **link budget equation** :

$$P_r|_{\text{dB}} = P_t|_{\text{dB}} + G_t|_{\text{dB}} + G_r|_{\text{dB}} - \underbrace{(L_p|_{\text{dB}} + \chi|_{\text{dB}})}_{\text{Pathloss function}}, \quad \xrightarrow{\text{better to measure it in dBm}} \quad (3.4)$$

It's somewhat like the Friis equation; we have the plus operation because we're in the log domain.

We can also rewrite it **more compactly as**

$$P_r|_{\text{dB}} = P_t|_{\text{dB}} + G|_{\text{dB}}, \quad (3.5)$$

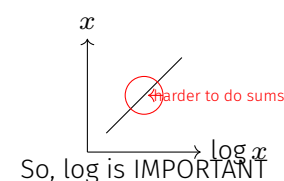
where **the gain**

$$G = \frac{G_t G_r}{L_p \chi} \quad (3.6)$$

Why they created logs?

In math, to "convert" products to sums.

$$\log(a \cdot b) = \log(a) + \log(b)$$



is the **large-scale channel gain**, a quantity which was already introduced in the signal processing section to normalize the channel response; only now it is acquiring its full significance. Of course if we have a log, the exponent can change.

10.4 Two-exponents model

We then have another **path loss modeling**; it gives the loss depending on the distance, which only applies after a certain reference distance D_{ref} . We have an expo-

ment η , which is normally higher than 2 and is called the pathloss exponent. We can also write this in dB:

$$L_p(D) = K_{\text{ref}} \left(\frac{D}{D_{\text{ref}}} \right)^\eta \quad (3.7)$$

$$L_p(D)|_{\text{dB}} = K_{\text{ref}}|_{\text{dB}} + 10\eta \log_{10} \left(\frac{D}{D_{\text{ref}}} \right) \quad D > D_{\text{ref}} \quad (3.8)$$

with D_{ref} being the reference distance and K_{ref} being the pathloss at D_{ref} .

This is a very rudimental model; we're only looking at the path loss.

Values of K_{ref}

If no value is specified for K_{ref} , it's safe to assume we can use the freespace value, as described in (3.9).

Channel modeling (pp. 131-208), page 8

The exponent η is the most consequential parameter, with typical values ranging between 3.5 and 4 if either the transmitter or the receiver are elevated. If both are within the propagation clutter, higher values can be encountered. Values below 2 are rare but possible in settings such as indoor corridors, outdoor urban canyons, or tunnels.

Channel modeling (pp. 131-208), page 8

The parameter K_{ref} is sometimes referred to as the intercept at D_{ref} because it is the value at which a plot of $L_p(D)$ intercepts a vertical axis drawn at D_{ref} . Given a reference distance, K_{ref} can be measured empirically. Alternatively, if the reference distance is chosen properly, K_{ref} can be computed under the assumption that the propagation from the antenna up to that point follows the free-space law described next. In this case, the representation in (3.8) becomes the two-slope model in Fig. 3.5 (see below here), namely **free-space decay** up to D_{ref} and decay **with exponent η thereafter**. This two-slope behavior can be justified analytically [51, section 2.4.1] and on the basis of experimental observations [213]. Besides the representation in (3.8), suitably complemented by log-normal shadow fading, several pathloss modeling alternatives are available. These alternatives are the subject of the remainder of this section. While the focus here is exclusively on how the large-scale propagation phenomena affect the local-average received power, models do exist for how other features of the local distribution (introduced later in this chapter) are affected, e.g., the Rice factor, the power delay profile, or the angle spread [214, 215].

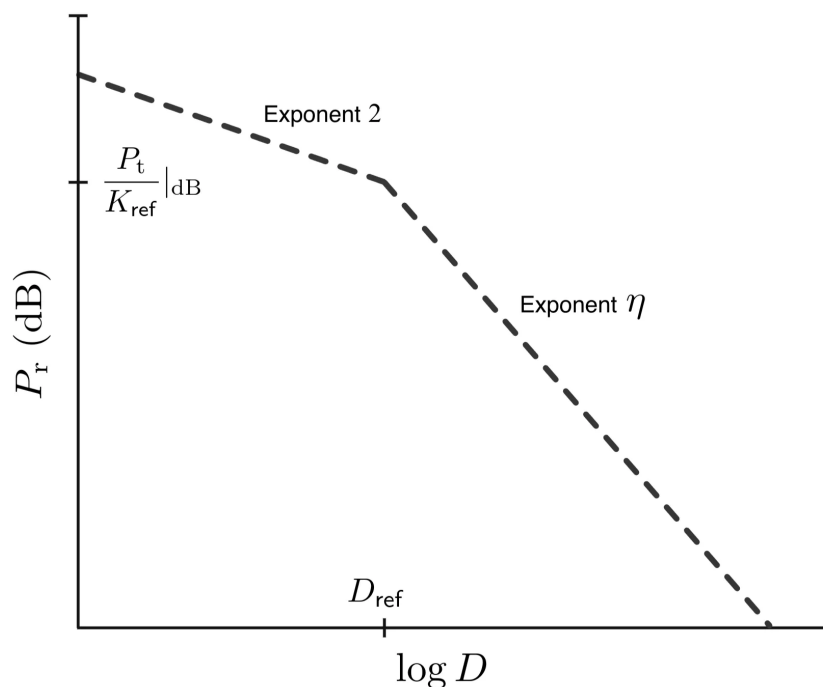


Fig 3.5

The first slope in the figure 3.5 is the **pathloss in free space**, given by

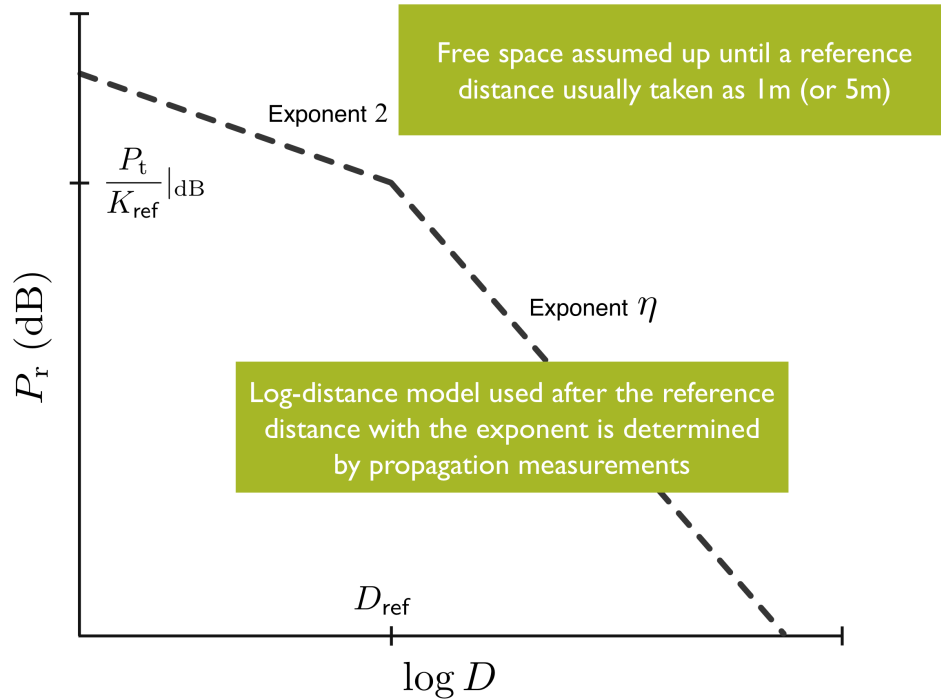
$$L_p(D)|_{\text{dB}} = 20 \log_{10} \left(\frac{4\pi D}{\lambda_c} \right), \quad (3.9)$$

which is none other than (3.8) adapted with $K_{\text{ref}} = 1$ and $\eta = 2$; λ_c is the carrier wavelength.

It **holds only when D lies in the far-field region of the antennas**, the rule of thumb being $D \geq 2d_a^2/\lambda_c$ where d_a is the largest physical dimension of either antenna. By definition, this model applies **only when there are no obstructions between transmitter and receiver or in the vicinity** of the trajectory linking them; this restricts its applicability to very specific situations, e.g., the computation of the intercept at a nearby point. There is no accompanying shadow fading. The presence of objects of any kind, even simply the ground, alters the square-law distance decay in (3.9), yielding a more general exponent η .

In class

We also have a two-slope model: up to a certain distance we have an exponent; then we have another exponent η :



If we look at the book, we see we have various models from [08.3_pp_131_208_Channel_modeling, page 8](#) to [08.3_pp_131_208_Channel_modeling, page 12](#).

10.5 Macrocell models

These are models with radius exceeding 1km.

10.5.1 Okumura-Hata model

This is a model derived by empirical measurements; it is

$$L_p|_{\text{dB}} = 69.55 + 26.16 \log_{10} f_c - 13.82 \log_{10} h_b - b(h_m) + (49.9 - 6.55 \log_{10} h_b) \log_{10} D \quad (3.10)$$

where:

- $f_c \in [150, 1500]$ is the carrier frequency in MHz
- $h_b \in [30, 200]$ is the base station antenna height in meters
- $h_m \in [1, 10]$ is the mobile user antenna height in meters
- D is in kilometers

and where

$$b(h_m) = \begin{cases} (1.1 \log_{10} f_c - 0.7)h_m - 1.56 \log_{10} f_c + 0.8 & \text{small to medium cities} \\ 8.29[\log_{10}(1.54h_m)]^2 - 1.1 & \text{large cities, } f_c \leq 300\text{MHz} \\ 3.2[\log_{10}(11.75h_m)]^2 - 4.97 & \text{large cities, } f_c \geq 300\text{MHz} \end{cases} \quad (3.11)$$

For **suburban areas** we should add to (3.10) the term:

$$-2[\log_{10}(f_c/28)]^2 - 5.4; \quad (3.12)$$

for **rural areas**, we should add to (3.10) the term:

$$-4.78(\log_{10} f_c)^2 + 18.33 \log_{10} f_c - 40.94. \quad (3.13)$$

This does not adapt to microcell; the COST-231 model adapts up to 2GHz.

10.5.2 COST-231 Hata model

This model works for frequencies from 150MHz to 2GHz. In urban areas, we have

$$L_p(D)|_{\text{dB}} = 46.3 + 33.9 \log_{10} f_c - 13.82 \log_{10} h_b - b(h_m) \\ + (44.9 - 6.55 \log_{10} h_b) \log_{10} D \quad (3.14)$$

with $b(h_m)$ just like in (3.11). In **dense metropolitan areas**, an extra 3dB must be added.

10.6 SUI model

This model works for fixed wireless antennas, like rooftop antennas, elevated of 15-40m over the clutter.

We have 3 types of terrains:

- A. Hilly terrains with heavy tree density
- B. Mostly flat terrains with moderate-to-heavy tree densities or hilly terrains with light tree densities.
- C. Flat terrains with light tree densities.

The main equation is a refinement of (3.8), aka

$$L_p(D)|_{\text{dB}} = K_{\text{ref}}|_{\text{dB}} + 10\eta \log_{10} \left(\frac{D}{D_{\text{ref}}} \right) + X(f_c) + X(h_m) \quad D > D_{\text{ref}}, \quad (10.1)$$

with $D_{\text{ref}} = 100\text{m}$ and $K_{\text{ref}}|_{\text{dB}}$ being the intercept at D_{ref} (assuming free space up to D_{ref}). We then have

$$\eta = a - bh_b + \frac{c}{h_b}, \quad (3.16)$$

with the constants being given according to the terrain type,

Table 3.1 Values of a, b, and c in (3.16)			
Parameter	Terrain A	Terrain B	Terrain C
a	4.6	4.0	3.6
b (m ⁻¹)	0.0075	0.0065	0.005
c (m)	12.6	17.1	20

h_m is the user height, which influences the model together with λ_c through:

$$X(f_c) = 6.0 \log_{10} \left(\frac{f_c}{2000} \right) \quad (3.17)$$

$$X(h_m) = \begin{cases} -10.8 \log_{10}(h_m/2000) & \text{Terrains A and B} \\ -20 \log_{10}(h_m/2000) & \text{Terrain C} \end{cases}. \quad (3.18)$$

10.7 Microcell models

Models with radius between 100m and 1km.

10.7.1 COST-231 Walfisch-Ikegami model

This model is defined for frequencies between 800MHz and 2GHz.

We define the model under the **3GPP parameters** (base station height $h_b = 12.5\text{m}$, building height 12m, building-to-building distance of 50m, street width of 25m, user height $h_m = 1.5\text{m}$).

We have that for **NLOS propagation**

$$L_p|_{\text{dB}} = -55.9 + 38 \log_{10} D + \left(24.5 + 1.5 \frac{f_c}{925}\right) \log_{10} f_c, \quad (3.19)$$

with f_c being in MHz, while D is in meters and must be $D \geq 20$.

Under **LOS propagation**,

$$L_p|_{\text{dB}} = -35.4 + 26 \log_{10} D + 20 \log_{10} f_c. \quad (3.20)$$

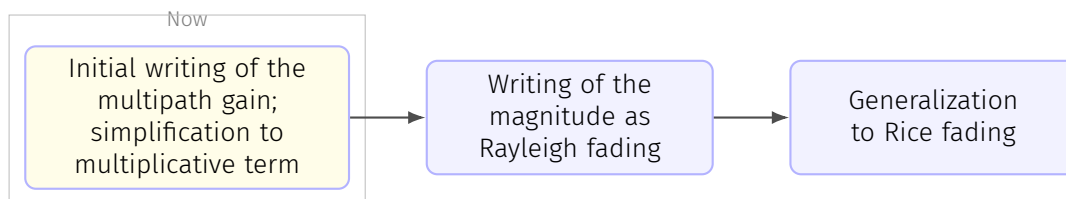
Here, the recommended standard deviation for shadow fading is $\sigma_{\text{dB}} = 4\text{dB}$. (?) End
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10.8 Picocell and indoor models

Generally we have a path loss like (3.8), with $\eta = 3.5$. The shadow fading is usually ranging between 3 and 6dB.

10.9 Small-scale fading

10.9.1 Multipath propagation (+Rayleigh and Rice fading)



Channel modeling (pp. 131-208), page 13

Let us now zoom in and enter the small-scale realm. When dealing with continuous-time representations, we use the complex **pseudo-baseband channel response**. We also recall from Section 2.2.3 that, **to discretize this representation**, lowpass filtering is in general required before sampling, although in those cases when the discretization returns a singletap response and the effect is only multiplicative, the complex gain of the pseudo-baseband channel can be borrowed directly.

Due to the various NLOS effects, what the receiver observes is a superposition of a number of replicas of the transmitted signal; the overall channel response is a sum of all these various contributions.

Each path can be described using the attenuation A_q and the delay τ_q , under the form

$$A_q \delta(\tau - \tau_q) e^{-j2\pi f_c \tau_q}.$$

 *Channel modeling (pp. 131-208), page 13*

Since they stem from objects within the local neighborhoods, all the relevant paths can be assumed to be subject to similar pathloss and shadow fading, suitably modified by the local reflection/diffraction/scattering, and thus **the complex gains $\{A_q\}$ may not exhibit major differences**. However, because at carrier frequencies of